

Arjun R Amarnath

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Summary

Computer Science graduate with hands on experience in full-stack development and AI/ML, with a focus on LLM powered applications and RAG systems. Experienced in Python, React.js, AWS, Ollama, and retrieval augmented generation, with experience building and deploying real-world web and AI applications.

Education

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| • Amrita Vishwa Vidyapeetham
B.Tech in Computer Science and Engineering
CGPA: 8.26 | Sept 2022 – Aug 2026 |
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Technical Skills

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| • Languages: Java, Python, SQL | • AI-ML & GenAI: YOLO, OpenCV, LLMs, RAG, Ollama |
| • Frontend: HTML, CSS, React.js | • Cloud Technologies: AWS (EC2, S3, API Gateway, CloudWatch, SNS), Docker |
| • Backend: Node.js, Express.js | • Version Control: Git, GitHub |
| • Database: MySQL, MongoDB | |

Professional Experience

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| Cue Court Coffee Full-Stack Developer | May 2026 – June 2026 |
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- Developed a full-stack booking platform with automated WhatsApp bookings, real time court blocking, and centralized slot tracking.
- Automated billing with court add ons, food orders, and membership-based pricing, reducing manual billing effort.
- Built CueBot, an AI powered analytics assistant for instant natural language insights into revenue, bookings, customers, and sales.

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| Perfect Study Space Full-Stack Developer | June 2026 – July 2026 |
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- Developed a multi-branch management platform supporting 3 branches, 5 staff, and 300+ student profiles, centralizing daily operations.
- Automated student registration, attendance, fee management, seat allocation, and notifications, reducing manual administrative work.
- Developed and deployed the organization website, increasing daily student enquiries by 50% across 3 branches.

Projects

AI-Powered Brain Tumor Detection & Clinical Decision Support System

- Developed a multimodal AI system for brain abnormality detection using CNN, EfficientNet, YOLO, and U-Net for MRI classification, localization, and segmentation.
- Integrated Grad-CAM for explainable AI and a local RAG-based GenAI assistant using Ollama to retrieve relevant context and generate natural-language clinical insights.
- Technologies Used: Python, PyTorch, CNN, EfficientNet, YOLO, U-Net, Grad-CAM, Ollama, RAG, OpenCV, Flask

Publications

8 papers accepted/published in Scopus-indexed international conferences (including IEEE and Springer).

1. QuickPark: Efficient Parking with Real-Time Pre-booking
2. Implementation of CPU Scheduling Algorithms Using Data Structures
3. SmartHoneypot: Honeypot System for Lightweight Cyber Threat Detection and Behavior Analysis
4. Validation of ARM Processor Instructions Using Deterministic Finite Automata

Extra Curricular Activities

President of Computer Society Of India (ASEB)

- Led the CSI team in organizing and executing technical events and coding competitions, overseeing task delegation and coordination while fostering effective teamwork and communication.